Cell Biology: Cell Cycle and Differentiation

Introduction

The production of the adult human is a choreographed process of cell proliferation and cell specialization. Throughout development cells rarely decide by themselves whether to divide or differentiate. In general, the two processes are negatively correlated with each other; cells which are increasingly differentiated, are less likely to divide; however, many fully differentiated cells are able to reenter the cell-cycle with appropriate stimulus. In this module, we look at the basic processes of division and differentiation and their governing mechanisms. When a cell stops listening and strikes out on its own independent path of division, cancer may result.

The number of closely related proteins that are involved in regulating the cell cycle is large and still increasing. Their interactions are complex and result in molecular circuits that are often difficult to fully understand. You are expected to know the major players and their interactions with other pieces of cellular machinery that affect their function. Our focus is on the basic machinery, but remember that there is a complex input into the cell-cycle machinery that may trigger or stop cell division and which may push the cells into a non-proliferative differentiation pathway. Although important, these inputs are not considered in this module.

Focus first on the basic steps and regulatory concepts in the cell cycle. The machine and its action have a basic activity, they are kinases. Slightly different versions of the machine are used at different stages of the cell cycle. The particular version of the machine in use executes the functions needed at that point in the cycle by through phosphorylation. Before acting, the machines often need to be activated by mechanisms that check whether the genome is in a healthy state (Checkpoints). Errors in the function of proteins involved in regulating the cell cycle and certifying the health of the genome are part of the development and progression of cancer. The other important activity of the machine is to activate the next version in the machine cell cycle and finally to self-inactivate. This pushes the cell cycle forward.

Reading

The Cell, 2nd ed: 571-603, 616-617

Need to know and understand

The stages in **the cell cycle** and characteristic events therein. **M, G0, G1, S, G2**
Points in the cell cycle: **G1 checkpoint, G2 checkpoint, M checkpoint, Start**
The molecular players regulating cell-cycle:

Cyclins (**Cycs**, **Clns, CLBs**), cell division cycle kinases (**cdc kinases**) cyclin-dependent protein kinases (**CDKs**) (These are a subset of the cdcs which work in a complex with the cyclins and are related in sequence to cdc2, which is therefore also known as Cdk1), **p53**, **p21**, retinoblastoma protein (**Rb**), Cdk inhibitors (CKIs), Sic1.

The role of **phosphorylation** and the **ubiquitin degredation pathway** (and its substrates marked with PEST sequences) in regulating the activity and amounts of cell cycle proteins at different stages of the cell-cycle.